VAMSI KRISHNA KOLLIPARA

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SUMMARY

Full-stack engineer who builds systems where being wrong is expensive: APIs other teams build on, delegated authorization over sensitive data, and LLM pipelines with guardrails around them. Currently the only engineer on client-facing products, owning them from discovery through production incident response. Earlier, backend and API work over multi-million-row utility billing and metering data.

SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Backend & APIs: Node.js, GraphQL, REST, serverless functions, PostgreSQL, SQL Server

AI / LLM: Anthropic Claude API, RAG, schema-validated structured output, LangChain, Pinecone

Cloud & Infra: AWS, Azure, Supabase, Vercel, GitHub Actions CI/CD

Frontend: React, Next.js, Tailwind CSS

EXPERIENCE

Full-Stack Engineer (Forward-Deployed) | Jul 2025 – Present
AI Data Management, Remote

Sole engineer on client-facing products in HIPAA-regulated environments; owned discovery through production operations.

- Designed and shipped a multi-tenant platform end-to-end: GraphQL API over Postgres with serverless functions, React frontend, GitHub Actions CI/CD deploying to Vercel and AWS. Consolidated HR/ATS, CRM, email, LMS, and e-commerce for 8 role types.
- Built the authorization model for delegated, scope-limited access: centralized RBAC across 8 roles, database-enforced Row Level Security, JWT-verified serverless endpoints, and Azure AD SSO. Every request had to prove who it acted on behalf of and what that principal could do.
- Owned a production LLM pipeline and the guardrails around it: retrieval, context management, schema-validated structured output so downstream code can trust the shape, per-user rate limiting, response caching, and human review before any output counted.
- Architected the chart-audit product behind that pipeline (Claude + RAG over EHR records, Azure Functions, Python, GraphQL, SQL Server), drafting a 120+ question clinical review protocol and raising physician throughput from 10 to 30+ audits/day.
- Built an idempotent ELT pipeline moving regulated data out of a third-party EHR into SQL Server on Azure: hourly batch jobs with concurrency control, structured logging, and failure alerting.
- Operated what I shipped: monitoring, production incident response, and usage-driven iteration, alongside discovery, technical proposals, and stakeholder demos.

Graduate Assistant | Aug 2023 – May 2025
Indiana State University, Terre Haute, IN

- Built a RAG knowledge assistant end-to-end, from ingestion through retrieval to generation, letting university staff query academic policy and HR procedure. LangChain, Pinecone, and a self-hosted open-source LLM over a curated 8,000-document corpus, with configurable semantic retrieval, chunk-overlap tuning, and vector indexing.
- Wrote and open-sourced WebWeaver, an async Python crawling library, and used it to ingest 40,000+ university URLs as that system's ingestion layer.
- Built internal web applications (React, Node.js, PostgreSQL) used by academic departments.

Programmer Analyst, Full-Stack | Jan 2022 – Jul 2023
Cognizant (Client: Duke Energy), Hyderabad, India

- Built and tested REST API endpoints serving metering, billing, and account data from Oracle and SAP systems to a high-traffic customer self-service platform; added caching and connection pooling that cut response times on the hottest endpoints.
- Optimized SQL against multi-million-row smart-meter (AMI) and billing tables: composite indexes, restructured joins, and eliminated N+1 patterns, cutting execution time on targeted queries by 30%.
- Built the billing and usage surfaces of the customer portal (usage analytics, billing history, outage status) with D3.js components; unit and integration tests (Jest, RTL) at 85%+ coverage.
- Built integration components and message handlers for the customer notification workflow (outage and billing alerts over email and SMS).

OPEN SOURCE (PyPI)

- **GitBar** (Mar 2026): macOS menubar Git dashboard aggregating pull requests, issues, and CI status across GitHub, GitLab, and Bitbucket. Python, PyObjC, REST APIs.
- **WebWeaver** (Sep 2024): async web crawling library with robots.txt compliance, deduplication, and BFS/recursive modes. Python, asyncio, aiohttp.

EDUCATION

M.S. Computer Science, Indiana State University | Aug 2023 – May 2025 | GPA 3.66

B.Tech Electronics & Communication Engineering, Vel Tech University, Chennai | Jun 2018 – May 2022 | GPA 9.01 / 10